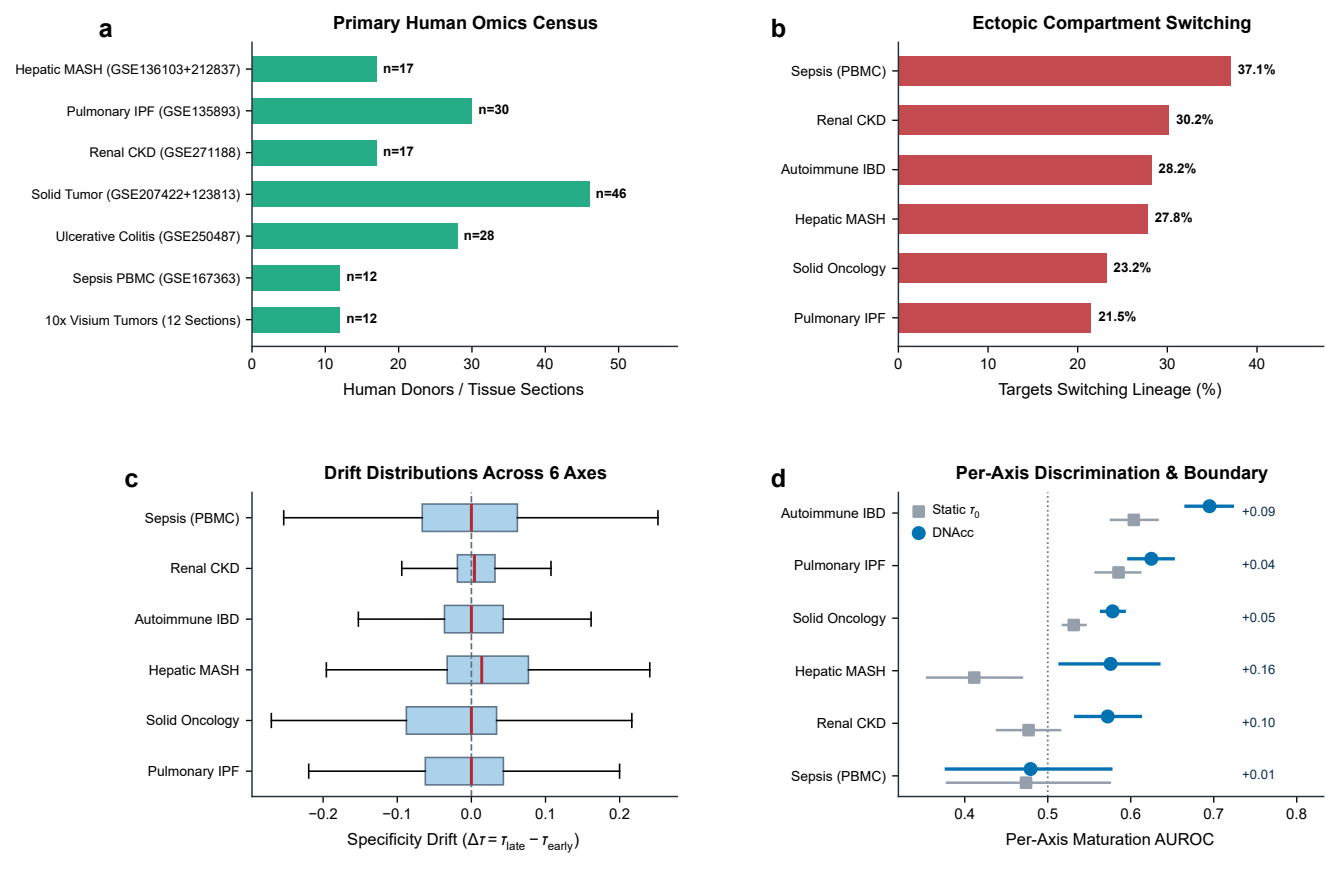

a
Primary Human Omics Census
Hepatic MASH (GSE136103+212837)
Pulmonary IPF (GSE135893)
Renal CKD (GSE271188)
Solid Tumor (GSE207422+123813)
Ulcerative Colitis (GSE250487)
Sepsis PBMC (GSE167363)
10x Visium Tumors (12 Sections)
n=17
n=30
n=17
n=46
n=28
n=12
n=12
0
10
20
30
40
50
Human Donors / Tissue Sections
b
Ectopic Compartment Switching
Sepsis (PBMC)
Renal CKD
Autoimmune IBD
Hepatic MASH
Solid Oncology
Pulmonary IPF
37.1%
30.2%
28.2%
27.8%
23.2%
21.5%
0
10
20
30
40
Targets Switching Lineage (%)
c
Drift Distributions Across 6 Axes
Sepsis (PBMC)
Renal CKD
Autoimmune IBD
Hepatic MASH
Solid Oncology
Pulmonary IPF
−0.2
−0.1
0.0
0.1
0.2
Specificity Drift ($\Delta\tau = \tau_{late} - \tau_{early}$)
d
Per-Axis Discrimination & Boundary
Static τ_0
DNAcc
Autoimmune IBD
Pulmonary IPF
Solid Oncology
Hepatic MASH
Renal CKD
Sepsis (PBMC)
+0.09
+0.04
+0.05
+0.16
+0.10
+0.01
0.4
0.5
0.6
0.7
0.8
Per-Axis Maturation AUROC